

Twin invariants: timestamp transitions • retain source/model/policy versions • immutable observations • simulation artifacts separate • preserve uncertainty • replay reproducible.

SEATRACE AI

Architecture • Workflow • Digital Twin • API

Overhauled implementation blueprint with the Maritime Digital Twin, visual workflow, evidence graph, complete REST endpoint inventory, technology decisions and production roadmap.

DESIGN PRINCIPLE

Evidence graph over a single score. Preserve uncertainty, competing hypotheses, immutable observations, provenance and replayable digital-twin state.

6. Data & Evidence Graph

Core entities: sar_scenes, detections, spills, drift_backtrack_runs, vessel_tracks, candidates, counterfactual_runs, digital_twin_states, evidence_events, cases, audit_log.

Evidence families: SAR • Environmental • Temporal • Origin/Drift • AIS • Counterfactual • Competing Hypotheses • Human Corroboration • Digital Twin State/Replay.

Central design: evidence graph is the source of truth; materialized incident/twin states are query-friendly snapshots derived from the event history.

7. REST API — Ingestion, Jobs & Analysis

Primary command/data contract.

Group	Method	Endpoint	Purpose
AUTH	POST	/api/v1/auth/login	Authenticate
AUTH	POST	/api/v1/auth/logout	End session
AUTH	POST	/api/v1/auth/refresh	Refresh token
AUTH	GET	/api/v1/auth/me	Current user
DATA	POST	/api/v1/data/sar/upload	Upload SAR
DATA	GET	/api/v1/data/sar/{id}	Scene metadata
DATA	GET	/api/v1/data/sar/search	Search scenes
DATA	GET	/api/v1/data/ais	AIS query
DATA	GET	/api/v1/data/weather	Weather
DATA	GET	/api/v1/data/ocean	Ocean
DATA	GET	/api/v1/data/context	Context
PROCESS	POST	/api/v1/process	Create async job
PROCESS	GET	/api/v1/process/{id}/status	Job status
PROCESS	POST	/api/v1/process/{id}/cancel	Cancel
ANALYSIS	POST	/api/v1/analyze/detect	Detection
ANALYSIS	POST	/api/v1/analyze/lookalike	Look-alike
ANALYSIS	POST	/api/v1/analyze/drift/backward	Backtrack
ANALYSIS	POST	/api/v1/analyze/drift/forward	Forward drift
ANALYSIS	GET	/api/v1/analyze/results/{id}	Result
ANALYSIS	GET	/api/v1/analyze/results/{id}/artifacts	Artifacts
ANALYSIS	GET	/api/v1/analyze/results/{id}/explanations	Explanations

8. REST API — Incidents, Vessels, Reports & Maps

Group	Method	Endpoint	Purpose
INCIDENTS	GET	/api/v1/incidents	List/search
INCIDENTS	POST	/api/v1/incidents	Create
INCIDENTS	GET	/api/v1/incidents/{id}	Details
INCIDENTS	PATCH	/api/v1/incidents/{id}	Update
INCIDENTS	GET	/api/v1/incidents/{id}/evidence	Evidence
INCIDENTS	GET	/api/v1/incidents/{id}/timeline	Timeline
INCIDENTS	GET	/api/v1/incidents/{id}/candidates	Candidates
INCIDENTS	POST	/api/v1/incidents/{id}/analyze	Analyze
INCIDENTS	POST	/api/v1/incidents/{id}/promote	Promote
INCIDENTS	POST	/api/v1/incidents/{id}/reject	Reject
INCIDENTS	POST	/api/v1/incidents/{id}/review	Review
INCIDENTS	GET	/api/v1/incidents/{id}/export	Export
VESSELS	GET	/api/v1/vessels/search	Search
VESSELS	GET	/api/v1/vessels/{mmsi}	Details
VESSELS	GET	/api/v1/vessels/{mmsi}/tracks	Track
VESSELS	GET	/api/v1/vessels/{mmsi}/gaps	AIS gaps
VESSELS	POST	/api/v1/vessels/{mmsi}/counterfactual	Counterfactual
VESSELS	GET	/api/v1/vessels/{mmsi}/evidence	Evidence
REPORTS	GET	/api/v1/reports	List
REPORTS	POST	/api/v1/reports/generate	Generate
REPORTS	GET	/api/v1/reports/{id}	Metadata
REPORTS	GET	/api/v1/reports/{id}/download	Download
REPORTS	POST	/api/v1/reports/{id}/regenerate	Regenerate
MAPS	GET	/api/v1/map/tiles/{z}/{x}/{y}	Tiles
MAPS	GET	/api/v1/map/layers	Layers
MAPS	GET	/api/v1/map/preview/{id}	Preview
MAPS	GET	/api/v1/map/incidents/{id}	Incident map

9. REST API — Digital Twin, Evidence & Operations

Group	Method	Endpoint	Purpose
TWIN	GET	/api/v1/twin/incidents/{id}	Current twin
TWIN	GET	/api/v1/twin/incidents/{id}/state	State snapshot
TWIN	GET	/api/v1/twin/incidents/{id}/timeline	Twin timeline
TWIN	GET	/api/v1/twin/incidents/{id}/replay	Replay metadata
TWIN	POST	/api/v1/twin/incidents/{id}/replay	Create replay job
TWIN	GET	/api/v1/twin/replays/{id}	Replay result
TWIN	POST	/api/v1/twin/incidents/{id}/what-if	Create hypothesis
TWIN	GET	/api/v1/twin/hypotheses/{id}	Hypothesis
TWIN	POST	/api/v1/twin/hypotheses/{id}/simulate	Run simulation
TWIN	GET	/api/v1/twin/incidents/{id}/layers	Twin layers
EVIDENCE	GET	/api/v1/evidence/{id}	Evidence item
EVIDENCE	GET	/api/v1/incidents/{id}/evidence/graph	Evidence graph
EVIDENCE	GET	/api/v1/incidents/{id}/evidence/gaps	Evidence gaps
EVIDENCE	POST	/api/v1/incidents/{id}/evidence/annotate	Annotate
AUDIT	GET	/api/v1/audit/incidents/{id}	Audit
MODELS	GET	/api/v1/models	Registry
MODELS	GET	/api/v1/models/{id}	Details
MODELS	GET	/api/v1/models/{id}/versions	Versions
USERS	GET	/api/v1/users	Users
USERS	POST	/api/v1/users	Create
USERS	GET	/api/v1/users/{id}	Details
USERS	PATCH	/api/v1/users/{id}	Update
USERS	DELETE	/api/v1/users/{id}	Delete
ORG	GET	/api/v1/organizations	Organizations
ORG	POST	/api/v1/organizations	Create org
ORG	GET	/api/v1/organizations/{id}	Org details
CONFIG	GET	/api/v1/config	Runtime config
CONFIG	GET	/api/v1/config/policies	Policies
OPS	GET	/api/v1/health	Health
OPS	GET	/api/v1/ready	Readiness
OPS	GET	/api/v1/metrics	Metrics
OPS	GET	/api/v1/version	Version

10. API Interaction & Future GraphQL Boundary

Current: REST for commands, uploads, jobs, map tiles, reports and integrations. SSE/WebSocket for live progress and twin updates.

Async: POST /api/v1/incidents/{id}/analyze → job_id → status/result endpoints; stream detection, drift, AIS, counterfactual and twin-update stages.

GraphQL later: add only if nested analyst reads justify it; REST commands remain primary.

11. Technology & Delivery

Fr Frontend React + TypeScript + MapLibre GL + deck.gl.	AP API FastAPI + REST + Pydantic.	As Async Celery + Redis.
AI AI/ML PyTorch + ConvNeXt-Tiny + DeepLabv3+ + XGBoost/LightGBM.	Ph Physics OpenDrift.	Ge Geospatial PostGIS + GDAL + Rasterio + Shapely + GeoPandas.
Tw Twin PostGIS temporal state + Twin State Manager + replay.	St Storage S3-compatible artifacts + MLflow.	Gr GraphQL Future query layer only if needed.
Go Go Future only if profiling proves need.	De Deploy Docker Compose first; scale after validation.	Ob Observability Structured logs and metrics.

12. Lifecycle, Validation & Roadmap

Ca Candidate Candidate → promoted incident using versioned policy.	An Analyze Detection → origin → AIS → counterfactual.	Tw Twin Synchronize observed and simulated state.
Re Review Analyst confirms, rejects, annotates or requests evidence.	At Attribution Candidate / unresolved / insufficient evidence.	Sc Scale Measured service separation and SaaS hardening.

Validation: multiple historical incidents; detection precision/recall, look-alike calibration, origin error, Top-1/3/5 ranking, counterfactual quality and twin replay reproducibility.